A New Control Method for Power-Assisted Wheelchair Based on the Surface Myoelectric Signal

Yuusuke Oonishi, Sehoon Oh, *Member, IEEE*, and Yoichi Hori, *Fellow, IEEE*

***Abstract*—In this paper, we design a new controller for power-assisted wheelchairs. Conventional power-assist controllers only focus on the amplification of the human input force using torque sensor. Our proposed controller does not use a torque sensor but an electromyogram sensor to estimate the driver's intention. We also use a disturbance observer to compensate for the lack of propelling torque information since it is difficult to simply regard the myoelectric signals as propelling torque. Then, we decide assist torque by combining the filtered myoelectric signals and the estimated human torque signal calculated by the disturbance observer. Experimental results show that the proposed power-assist method realizes fast, stable, and smooth acceleration.**

***Index Terms*—Disturbance observer, electromyogram (EMG), power-assisting control, wheelchair.**

I. Introduction

According to demographic forecasts in Japan, the ratio of elderly people at least 65 years old is expected to top 40% in 2050. Therefore, there are needs for technical assistance for elderly people from the aspect of welfare engineering based on the coming aging society with fewer children. We focus on power-assisted wheelchairs (shown in Fig. 1), which are self-propelled wheelchairs with an electrical motor. There are two advantages of using power-assisted wheelchairs. First, they reduce fatigue when going long distances. Second, they aid in maintaining the present abilities of the operators as well as rehabilitation. In other words, power-assisted wheelchairs have advantages of both self-propelled wheelchairs and electric-powered wheelchairs. For these reasons, power-assisted wheelchairs have become widespread. But current power-assisted wheelchairs on the market only amplify the power through the handrims and assist, leaving much to be improved. Therefore, various assist methods have been proposed to provide smoother operation [1]–[5].

In this paper, we do not use any torque sensor [6] or joystick [7]; rather, we use an electromyogram (EMG) sensor for input interface, considering developing cooperative control between human and machine. Ideally, the magnitude of an EMG sensor attached to a muscle should be proportional to the force exerted by the muscle. However, it is not so; the magnitude of the EMG sensor output differs depending on people, and even for the same person, the magnitude becomes different depending on the fatigue of muscle and the conditions of the skin the EMG sensor is located on [8]. Thus, it is difficult to simply consider that input torque and myoelectric signal are proportional. For this reason, we introduce the disturbance observer to detect the exerted human force and to determine how much torque should be provided by motors to assist the user. Moreover, we combine the EMG signal with the estimated force to provide more safe and comfortable assistance.

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Fig. 1. Power-assisted wheelchair as an experimental tool (YAMAHA JW II).

II. Analysis of Conventional Method of Power-Assist Wheelchair

A. Fundamentals of Conventional Method

In this section, we briefly present a conventional power-assist method, which has been implemented for most of the commercial-type power-assisted wheelchairs. The conventional control method of power-assisted wheelchairs simply amplifies the manual inputs from the push handrims with first-order delay. The equation of the power-assist controller is described as

$$T_{\text{assist}} = K\frac{1}{1+\tau s}T_{\text{human}} \tag{1}$$

where K is the power-assist-ratio, T_{human} is the input torque from the push handrim, T_{assist} is the amplified torque from the push handrim, and τ is the time constant of the first-order delay. τ should be a suitable value to realize an inertia for the wheelchair. Therefore, τ should be a small value when the differential value of the input torque is positive (which means

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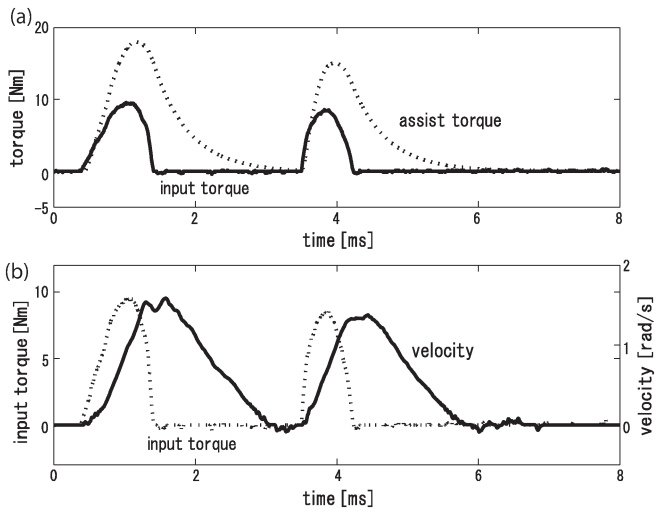

Fig. 2. Power-assist result of the conventional method.

the wheelchair is accelerated) and τ should be a large value when the differential value of the input torque is negative (which means the wheelchair is decelerated), as shown by the following relations:

$$\tau = \begin{cases} \tau_{\text{fast}}, & (\dot{T}_{\text{human}} > 0) \\ \tau_{\text{slow}}, & (\dot{T}_{\text{human}} < 0). \end{cases} \tag{2}$$

For example, our experiments adopt the following values, respectively:

$$\tau_{\text{fast}} = 0.08\ [\text{s}] \quad \tau_{\text{slow}} = 1.0\ [\text{s}]. \tag{3}$$

B. Experimental Result of Conventional Power-Assist Method

Experiments were carried out with real-time control using a YAMAHA JWII wheelchair and ARTLinux PC, with a torque sensor, a gyro sensor, and two rotary encoders. The behavior of this controller is shown in Fig. 2. The result shows that there are two possible problems to be improved in the acceleration phase. First, the rising time of the wheel speed is long. Second, the acceleration is too sudden. These problems tend to cause driver discomfort and dangerous feeling. Therefore, we solve these problems by using a myoelectric sensor rather than a conventional torque sensor.

III. Signal Processing of EMG

A. EMG Signal

EMG records action potential of the muscle fiber, which causes contraction of the muscles. EMG is established as an evaluation tool for applied medical rehabilitation, physical education, ergonomics, and so on [9]–[13]. There are many engineering applications such as myoelectric switch [14], myoelectric hand [15], robot suit [16], and wheelchair [17], [18]. The advantages of introducing an electromyography signal for the power-assist control of the wheelchair are as follows:

1) able to assist people who do not have enough power to propel handrims;

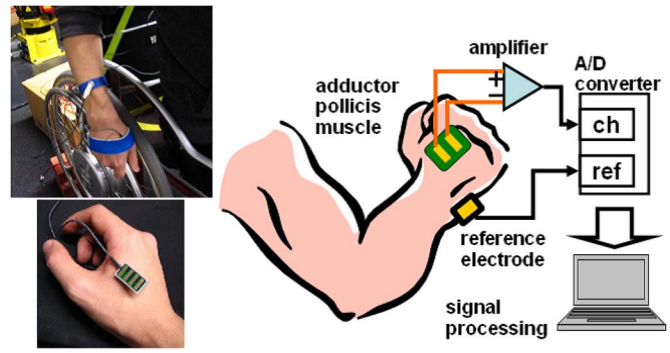

Fig. 3. Measurement system of the EMG signals.

2) able to detect how much power is applied;
3) able to estimate the driver's motion.

First, by using an EMG sensor, we expect power assistance and also the effect of rehabilitation: Elderly people with little grip force or having difficulty in propelling handrims because of a disorder of the hands can drive a wheelchair by themselves.

Second, a myoelectric signal enables measurement of how much power is applied by the operator.

Third, a myoelectric signal measures a signal that occurs when the brain orders the muscle to propel the wheelchairs before the wheels rotate.

B. EMG Sensor

There are many kinds of EMG sensor, but we use a dry-surface electrode [19] as it is noninvasive and easy to apply. As shown in Fig. 3, we carried out measurement by applying 2.5 V to a reference electrode and having a surface electrode attached to the adductor pollicis muscle. As in Fig. 3, EMG measured by the surface electrode on the skin is called surface electromyogram (sEMG). It is suitable for measuring entire muscle activation because it is a temporal and spatial summation of the action potential of the muscle fiber under the surface electrode. The sMEG is 1 200 times amplified by the differential amplifier as the raw EMG signals are between a few microvolt and 2–3 millivolt. The sampling rate is 1 kHz through a 12-bit A/D converter. The human subject was a 23-year-old healthy man.

C. Signal Processing of EMG

The raw signals of EMG and encoder signals when a driver propels the handrims twice are shown in Fig. 4. The high-frequency EMG signals are detected before the driver propels the handrims. This is an advantage of the EMG sensor; human's will can be detected very quickly. Here, we define the grasping phase as the range of about 0.5 s before and after propelling the handrims. We also define the propelling phase as the range during which the driver propels the handrims and wheel velocity is increasing.

To extract the grasping phase from EMG signals, it is necessary to apply signal processing to the EMG signals of the adductor pollicis muscle. A low-pass filter (LPF) is frequently adopted for filtering of the EMG signals. LPF is frequently adopted for filtering of the EMG signals. This LPF, however,

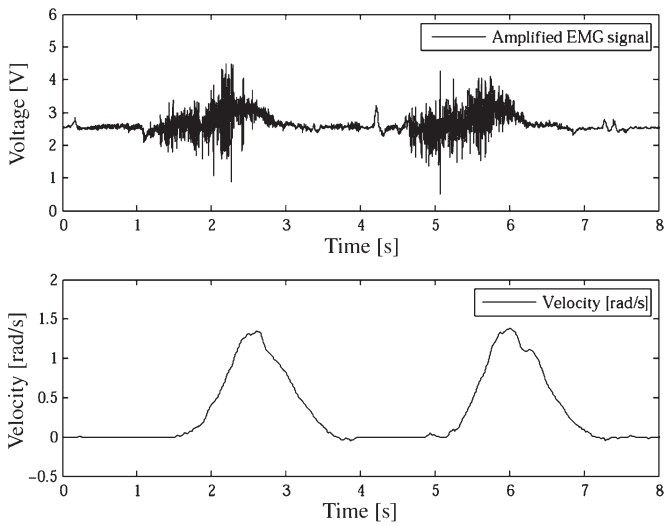

Fig. 4. Amplified EMG signals of the adductor pollicis muscle and the wheel velocity of the wheelchair.

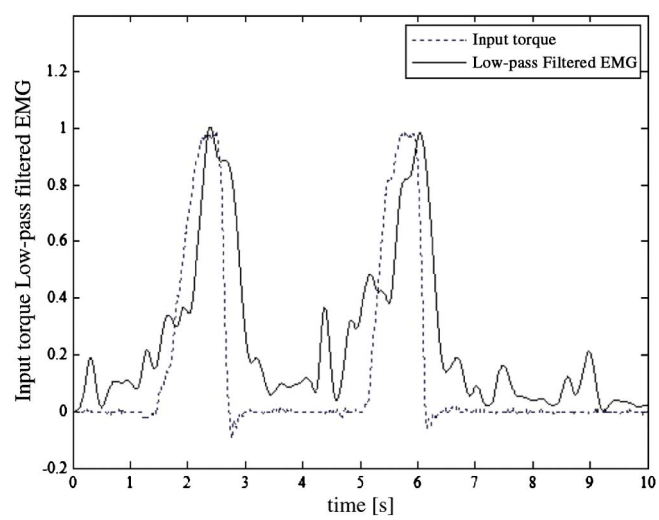

Fig. 5. Comparison between low-pass filtered EMG signal and input torque via the torque sensor.

is not adequate to detect the grasping phase. Fig. 5 is one result of low-pass filtering of EMG signals with a cutoff frequency of 3 Hz. The EMG signal is smoothed, and due to this smoothing, the signals representing the grasping phase are attenuated; thus, it is difficult to point out the starting point of the grasping phase.

Therefore, the sEMG signals are digitally high-pass filtered as in the following equation:

$$\mathrm{EMG}_{\mathrm{hpf}}[k] = \frac{1}{1+\omega T}\left(\mathrm{EMG}_{\mathrm{hpf}}[k-1] + \mathrm{sEMG}[k] - \mathrm{sEMG}[k-1]\right) \quad (4)$$

where ω is the cutoff frequency of 100 rad/sec, T is the sampling time of 1 ms, and $\mathrm{EMG}_{\mathrm{hpf}}[k]$ is the high-pass filtered EMG signal. Next, the signals are full-wave rectified. Then, the

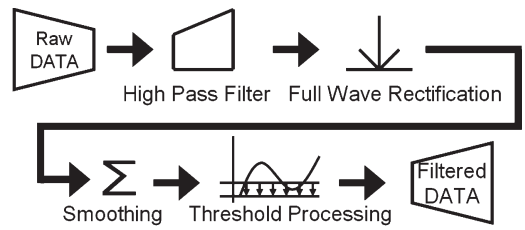

Fig. 6. Flow diagram of EMG signal processing.

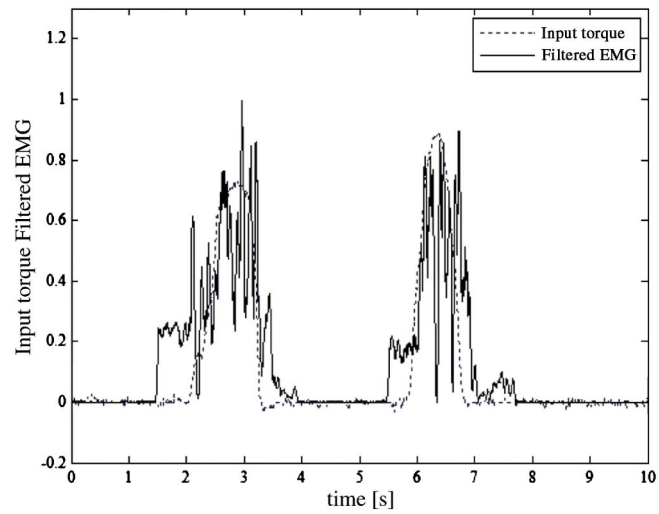

Fig. 7. Comparison between filtered EMG signal and input torque via the torque sensor.

signals are smoothed by taking a moving average per 50 points as in the following equation:

$$\mathrm{EMG}_{\mathrm{filtered}}[k] = \frac{1}{50}\sum_{i=0}^{49}\left|\mathrm{EMG}_{\mathrm{hpf}}[k-i]\right|. \quad (5)$$

This process is effective in reducing the impulsive noise that is generated by the disconnection between the human skin and the EMG sensor. Finally, subthreshold signals are cut to zero so that the steady noise is eliminated. The signal processing of EMG signals is shown in Fig. 6.

The filtered EMG signal of the adductor pollicis muscle is shown in Fig. 7. Compared with input torque signals from the torque sensor, there is a large rise in the filtered EMG signal before the rise of the measured input torque, which means the suggested filtering captures the grasping phase successfully. Therefore, EMG signals of the grasping phase can be used for starting or switching on of the assistance to prevent a jerky assisting force. Then, the main assisting torque is controlled by the estimated human torque.

IV. Acceleration Controller of a Power-Assist Wheelchair

A. Estimation of the Propelling Torque

At the propelling phase, the main assisting torque is determined by the estimated human torque. Therefore, we calculate assist torque by using a disturbance observer [20] at the propelling phase. The block diagram of the disturbance observer is

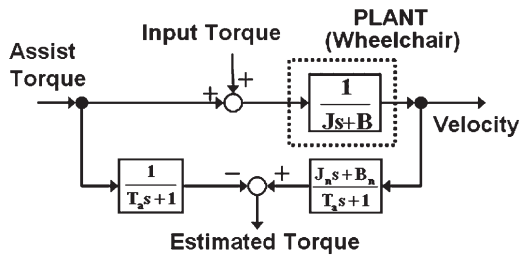

Fig. 8. Block diagram of the disturbance observer.

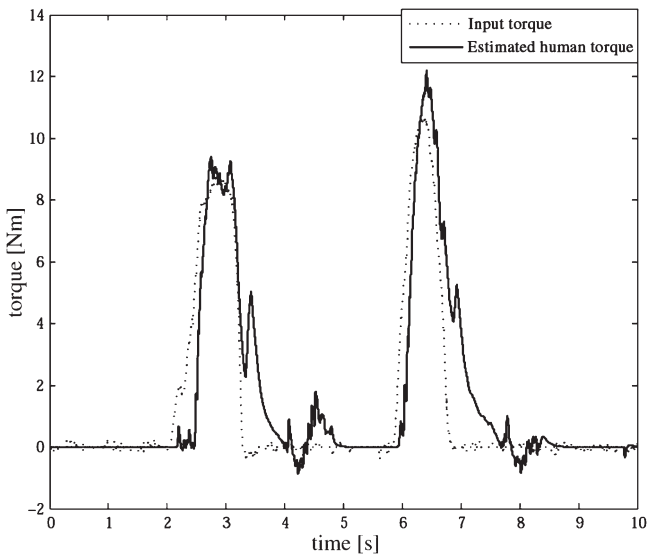

Fig. 9. Estimation result of the human force calculated by the disturbance observer.

shown in Fig. 8, where J_n is the nominal value of J, B_n is the nominal value of B, and T_a is the time constant of the LPF. We can estimate human input torque by calculating the difference between the commanded value of the assist torque and the inverse plant of the nominal model. The experimental result is shown in Fig. 9. Compared with input torque signals from the torque sensor, estimated torque signals from the disturbance observer can estimate human input torque at the propelling phase, although the disturbance observer does not work at the grasping phase.

B. Design of the Power-Assist Controller

At the grasping phase, we regard filtered EMG signals as the intensity of the driver's intention of forward movement, and at the propelling phase, we estimate propelling torque based on the disturbance observer. Then, we combine filtered EMG signals and estimated torque signals using a weighting function of velocity. The illustration of the whole algorithm is shown in Fig. 10. K_1 and K_2 are constant values and F_1 and F_2 are variables that indicate the effect extent of power assistance. These values change according to the wheel velocity. At the grasping phase, when the wheel velocity is under v_α, we set F_1 to a large value and set F_2 to a small value so that power assistance is based on the filtered EMG signals. On the other hand, at the propelling phase, when the wheel velocity is over v_β, we set F_1 to a small value and set F_2 to a large value so that power assistance is decided based on the estimated torque

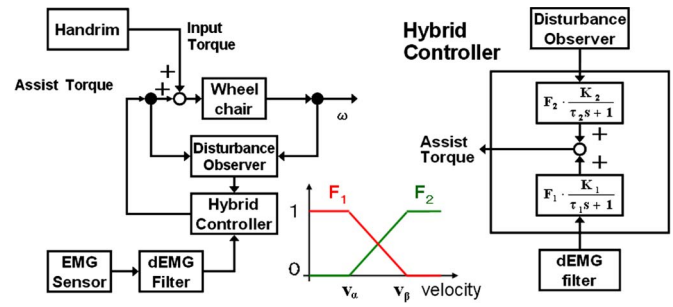

Fig. 10. Block diagram of the acceleration controller.

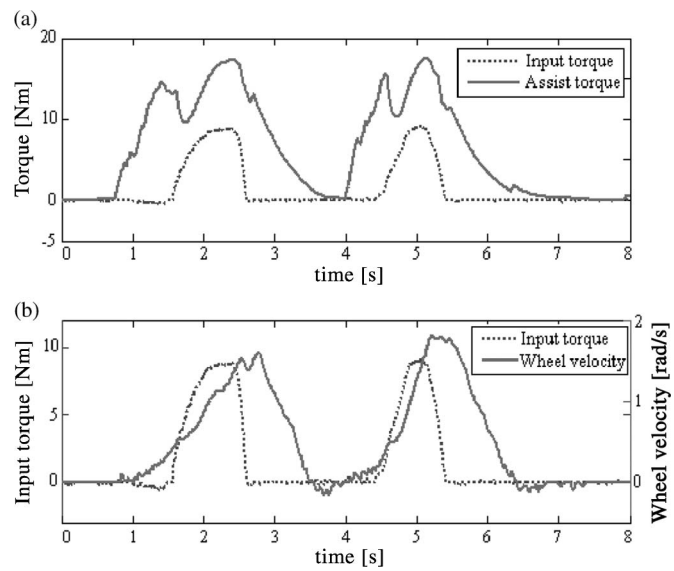

Fig. 11. Power-assist result of the proposed power-assist method (in the forward direction) shown in Fig. 10 when the driver propels twice at the forward direction.

signals. At the transition phase from grasping to propelling, F_1 decreases and F_2 increases in proportion to acceleration to realize smooth acceleration. At the coasting phase, we use the same method as the conventional method. v_α and v_β are used as parameters, which indicate the effect between filtered EMG signals and estimated torque signals. For example, we set v_α to a large value when the driver is a severely disabled person who has a weak grip. When we use the proposed method for rehabilitation, we gradually reduce v_β according to the degree of recovery. In this way, we can realize acceleration according to the driver's taste by changing v_α and v_β.

C. Experiments

Experiments where the human subject propels handrims two times using the proposed acceleration method are performed. The results are shown in Fig. 11. Comparing Fig. 11(a) with Fig. 2(a), we can see that the assist torque of the proposed method is generated earlier than that of the conventional method. Shown in Fig. 11(b), wheel velocity accelerates smoothly before the driver propels the handrims so that the driver can propel with less power. When the wheelchair runs at the dirt road or lawn, it may need strong power for the driver to propel the wheelchair. The proposed method has a beneficial effect on these situations.

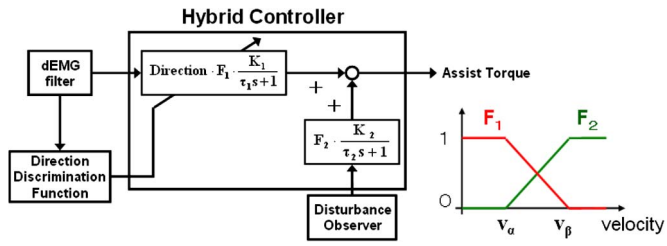

Fig. 12. Internal structure of a hybrid controller block.

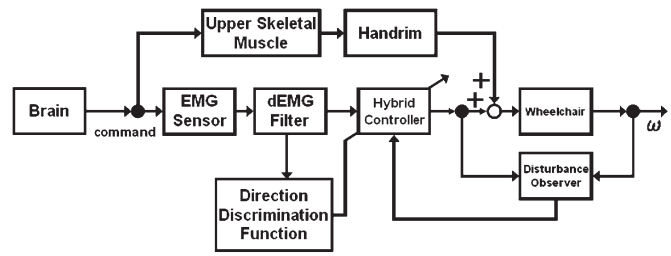

Fig. 13. Block diagram of the proposed power-assist method.

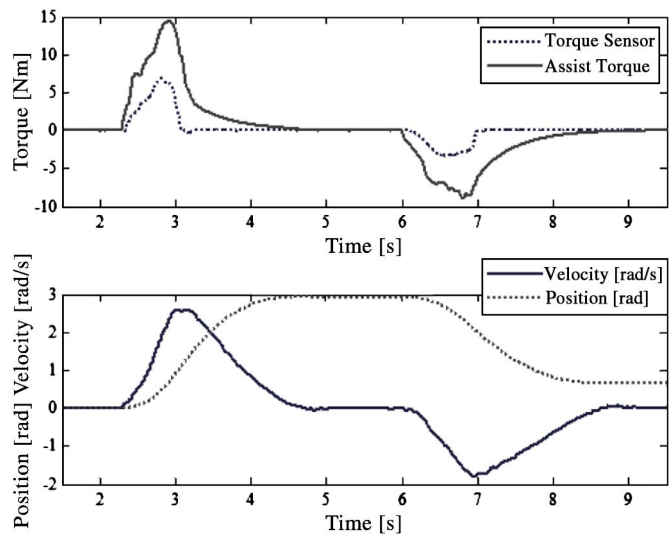

Fig. 14. Power-assist result of the proposed power-assist method (in two directions) shown in Fig. 13 when the driver propels at the forward and backward direction, respectively.

V. Proposed Power-Assistance Method

In Section IV, we proposed the acceleration controller of the power-assisted wheelchair by using EMG sensor and disturbance observer. However, the proposed controller does not work in the backward direction, when the driver's intention is backward movement. In this section, therefore, we suggest a new direction-detection algorithm, which can assist the wheelchair in both forward and backward direction.

A. Design of Proposed Power-Assist Controller

Since a driver uses the entire upper extremity at the forward and backward movement, it is difficult to estimate whether the driver wants to propel forward or backward through the filtered EMG signal of the adductor pollicis muscle. At forward movement, a driver flexes his or her arms and then extends his or her arms. On the other hand, the driver extends his or her arms and then flexes his or her arms at backward movement. In consideration of the aspect of this driver's movement, we focus on the flexor muscle and the extensor muscle of the upper arm to determine the direction. We attached a surface electrode to the triceps brachii muscle and the biceps brachii muscle. Then, we add a direction-discrimination function to the acceleration controller in Fig. 10. In Fig. 12, the direction is controlled by the sign of the torque, which is decided by the direction-discrimination function. The direction-discrimination function makes positive torque by a positive gain when filtered EMG signals of the biceps brachii muscle is stronger than those of the triceps brachii muscle at the beginning time of the grasping phase. On the other hand, when the filtered EMG signals of the triceps brachii muscle is stronger than those of the biceps brachii muscle at the beginning time of the grasping phase, the direction discrimination function makes the assist-torque negative.

B. Block Diagram of Proposed Power-Assist Method and Experimental Result

The integrated block diagram of the proposed power-assist method is shown in Fig. 13. In Fig. 13, input torque is added to the handrim by the command of the driver through the upper skeletal muscle. The disturbance observer estimates this input torque as the estimated torque signals and carries these signals to the hybrid controller as the feedback signal. At the same time, the dEMG filter estimates filtered EMG signals as the intensity of driver's intention of propelling movement and carries these signals to the hybrid controller as the feedforward signal. In addition, the direction-discrimination function estimates whether the driver wants to propel forward or backward. Then, the hybrid controller combines feedforward and feedback signals and decides the assist torque.

Experiments where the human subject propels handrims forward and backward, respectively, using the proposed power-assist method are performed. The results are shown in Fig. 14. Fig. 14 shows that the proposed power-assist method can assist the wheelchair smoothly in both forward and backward direction. Here, we set K_1 and K_2 smaller at the backward movement than at the forward movement since too much assist in the backward direction cause dangerous situations such as rollover of the driver.

VI. Conclusion

In this paper, we have designed a new controller for the power-assisted wheelchair. We assist the wheelchair by combining filtered EMG signal and estimated torque signals calculated by the disturbance observer. At the grasping phase, we assist the wheelchair by using the filtered EMG signal so that the wheelchair can move before the driver propels the handrims. The power-assisted direction is decided depending on the relative strength of filtered EMG signals between the triceps brachii muscle and the biceps brachii muscle. This lightens the driver's burden when he or she starts to propel the handrims. At the propelling phase, we assist the wheelchair by using the estimated torque signals so that a caregiver's pushing force can be also amplified. This, on the other hand, lightens the caregiver's burden.

It also provides acceleration according to the driver's desired specification by changing K_1, K_2, v_α, and v_β of Fig. 12.

From the viewpoint of the welfare application, smooth control and ride quality are important. More detailed discussion on the analysis of the control system such as the tradeoff between assistance performance and weak robustness should be researched in future work. Experiments for many test subjects and subject rating are also left for further investigation.

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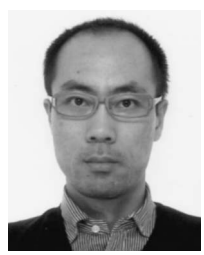

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